CSE3002 –AI – Digital Assignment -1

1. Identify the types of agents for the following. Also, describe the PEAS description for the following.

   a) Room cleaner
   b) Self-Driving Cars
   c) Robo-Chitti
   d) Tic-Tac-Toe playing agent

2. Trace the constraint satisfaction procedure, solving the following crypt arithmetic problem:

```
   B A S E
 + B A L L
 ----------
 G A M E S
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```
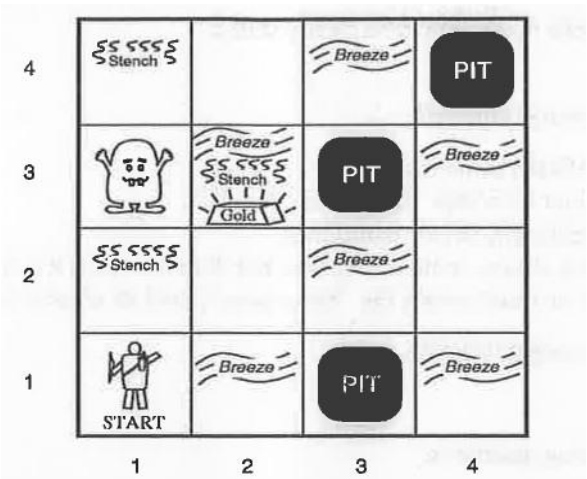
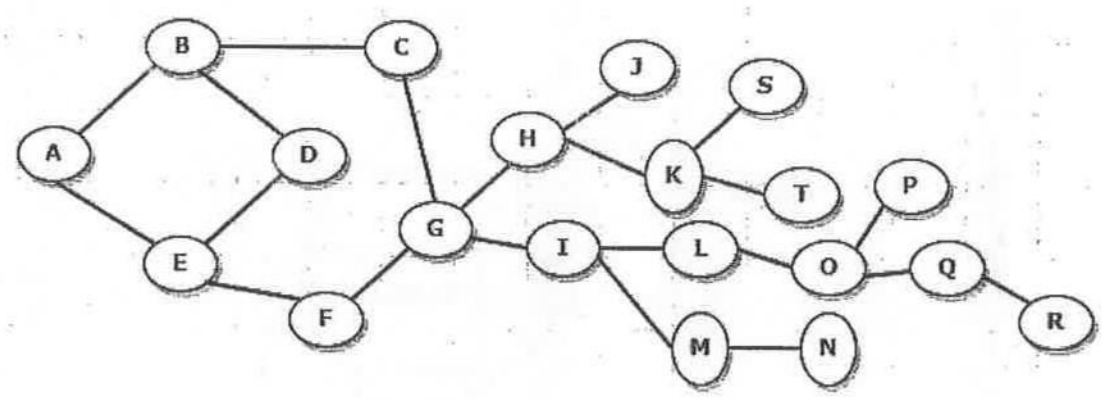

3. The Wumpus world is a cave with 16 room (4×4). Each room in connected to others through walkways (no rooms are connected diagonally). The knowledge-based agent starts from Room [1, 1]. The cave contains some pits, a treasure, and a beast named Wumpus. The Wampus cannot move but will eat anyone who enters its room. If the agent enters a pit, it gets stuck there. The goal of the agent is to take the treasure and exit the cave. The agent is rewarded when the goal conditions are met and penalized if it falls into a pit or is eaten by the Wumpus. Certain elements assist the agent in exploring the cave: the rooms adjacent to the Wumpus are filled with a stench, the adjacent rooms of pits are breezy, and the treasure room is glittery. The agent is given one arrow, which it can use to kill the Wumpus when facing it (the Wumpus screams when killed).

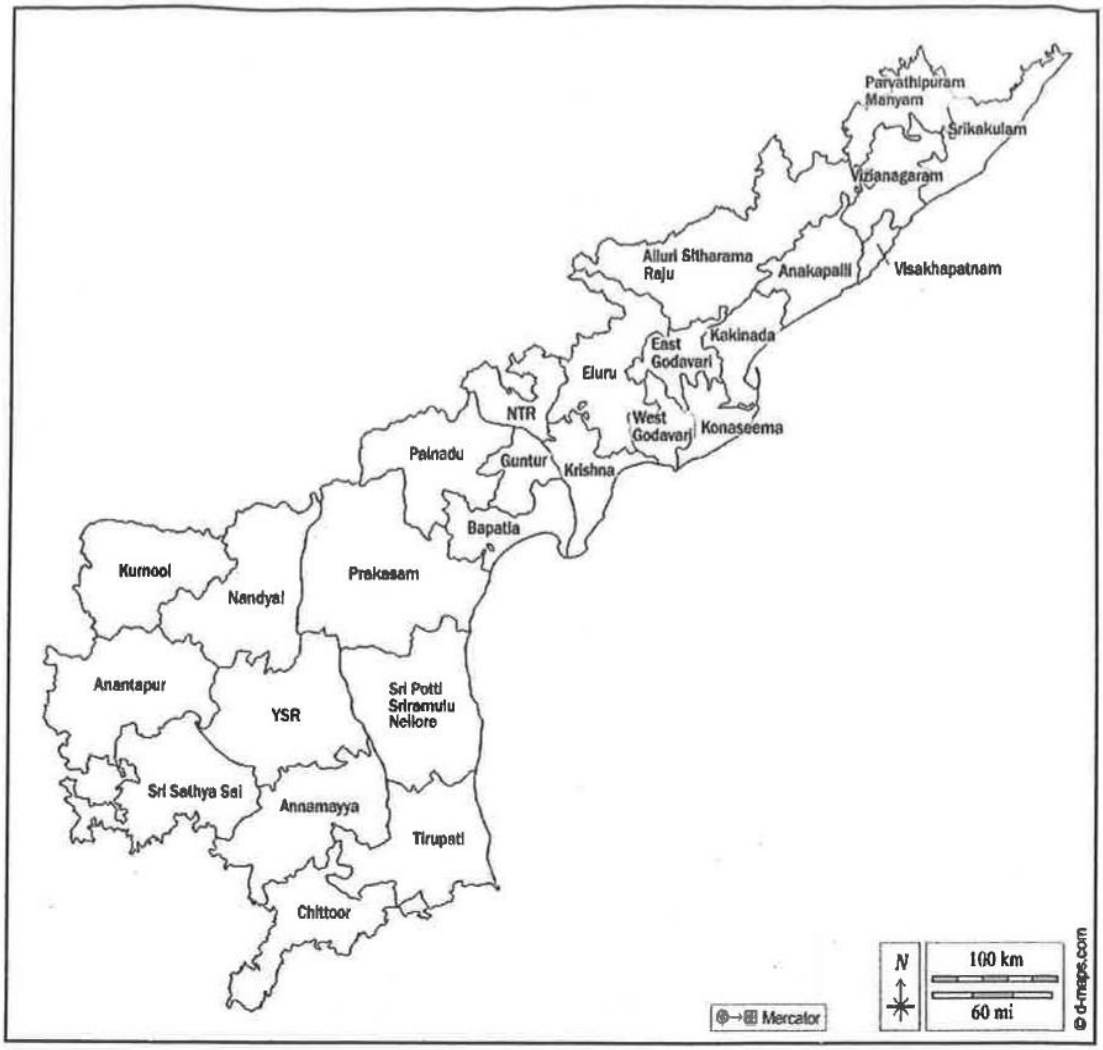


provide the PEAS Description for the Wumpus World problem and prove that the Wumpus Is in Room (1,3) using the necessary rules of inference.

4. Consider the following graph; **find the traversal order to reach the solution node T** from starting **node B**. Solve the problem **using both DFS and BFS**.



5. Use backtracking **to color the map of Andhra Pradesh** given below following the constraint that no two adjacent cities should have the same colours
(**Find the chromatic number for the below map**).



6. Given an initial state of the **8-puzzle problem** and final state to be reached, find the most cost-effective path to reach the final state from initial state using A* Algorithm. Consider **g(n)=Depth of node** and **h(n)=Number of misplaced tiles.**

1	2	3
4	8	
7	6	5

Initial State

1	2	3
4	5	6
7	8	

Final State

7. Explain the following Topics with neat Diagram.

   i) **Approaches in Knowledge representation.**

   ii) **Types of Agents.**

8. What is Artificial Intelligence? Discuss the **History of AI and List out the Applications of AI.**